BPM/linearity development: paired conversion gain increment

Date: 3 November 2026. Status: exact numerical interface proposed for owner review. Base: c73d9a9e3a631fbbb2346beaa3c150c110a60343. This advances the accepted 2 November formula design; no new numerical source is implemented by this review. Existing qualified bias/BPM execution remains available.

Current owner decisions

- Continue BPM/linearity recipe development. QC investigation and visualization remain deferred.
- The eventual linearity correction is applied in mefpp. Later validation reduces the SAME science frames with and without that correction and compares them in the photometric calibration notebook. That experiment is not this increment.
- Use the selected Mac OmegaCAM sequence below. Nonconsecutive members are paired explicitly; chronological order and acquisition times remain unchanged.
- Controlled lamps support testing small pair mismatches; do not infer perfectly constant illumination or introduce unrestricted per-exposure lamp parameters.
- Keep the unsigned 8-bit reason BPM, independent coverage, reusable Eclipse numerical services, and the completed bias recipe orchestration template.

Source reuse review

Reviewed main's Eclipse sources and the existing BPM service:

Existing resource	Actual role	Reuse decision
src/iproc/image_stats.c: image_getstats_masked_checked	Finite, BPM-selected ROI mean, median, sample standard deviation and optional MAD	Reuse for four-frame means and counts
src/iproc/image_stats.c: image_pair_stats_checked	Pair difference, intersected maps, checked regional statistics	Reuse for flat and RAW-bias differences
src/iproc/detector.c: cube_create_gainmap_robust / proportional	Relative pixel response against measured frame level	Retain for response BPM; not conversion gain in electrons/ADU
src/iproc/detector.c: detector_linearity_fit	Per-pixel time polynomial in signal, no constant term; diagnostic is mean squared time residual	Preserve legacy API; insufficient for the shared signed shutter model
src/iproc/detector.c: cube_correct_detlin	In-place cubic using b/a and c/a	Potential later correction primitive; explicit coefficient conversion and validity checks required
src/math/fit_curve.c and matrix.c	Polynomial/linear fits and normal-equation least squares	Retain scaffolding; no joint constrained covariance fit is supplied

The current BPM service already handles active ROIs and explicit sample maps. File admission already carries pair identifiers. No replacement FITS reader, statistics implementation, BPM response fitter or acquisition-order sorting is needed.

Measurements and notation

The calculation uses **two flat-field exposures and two bias exposures** from the same detector and readout configuration. A pair means two exposures at the same integration time; they need not be consecutive in the acquisition sequence.

An ROI (region of interest) is the rectangular detector area being measured. The **same accepted pixels** must be used for every mean and difference below. A pixel is accepted only when selected by the BPM and finite in all four images. The recipe also excludes samples outside the explicitly selected saturation and signal limits. There is no automatic clipping.

Symbol	Meaning	Unit
p	Position of a pixel within the ROI	Pixel coordinate
G	Set of accepted pixel positions, common to all four images	—
N	Number of pixels in G; at least two are needed for a variance	Pixels
$F_{1,p}$, $F_{2,p}$	Raw values at pixel p in the first and second flat exposures	ADU
$B_{1,p}$, $B_{2,p}$	Raw values at pixel p in the first and second bias exposures	ADU
z_p	Any image or difference value at pixel p	ADU
$\bar{z}$	Arithmetic mean of z_p over G	ADU
$v(z)$	Sample variance of z_p over G, with denominator $N-1$	ADU^2
ΔF_p	Flat-pair difference, $F_{1,p} - F_{2,p}$	ADU
ΔB_p	Bias-pair difference, $B_{1,p} - B_{2,p}$	ADU
g	Conversion gain: number of electrons corresponding to one ADU	electrons/ADU
$\mathrm{RON}_{\mathrm{ADU}}$	Readout noise of one exposure, expressed in ADU	ADU
$\mathrm{MAD}(z)$	Median absolute deviation from the median of z_p, over G	ADU

The subscript in $\mathrm{RON}_{\mathrm{ADU}}$ is the complete unit label "ADU"; it is not a product of variables. Bars above image names, such as $\bar{F}_1$, mean averages over the common set G.

Formulas to implement

Mean and sample variance

For any image or difference z:

$$\bar{z} = \frac{1}{N}\sum_{p\in G} z_p, \qquad v(z) = \frac{1}{N-1}\sum_{p\in G}(z_p - \bar{z})^2.$$

The variance measures the scatter after removing the arithmetic mean. All means and variances below use this same definition and pixel population.

Readout noise from the raw bias pair

Form the difference of the two raw biases:

$$\Delta B_p = B_{1,p} - B_{2,p}.$$

For independent exposures with equal readout-noise variance, the difference contains twice the variance of one exposure. Therefore:

$$\boxed{\mathrm{RON}_{\mathrm{ADU}} = \sqrt{\frac{v(\Delta B)}{2}}}.$$

No overscan correction is applied before this subtraction, preserving the previously agreed raw-pair measurement without scan-correction correlations.

The optional robust diagnostic is:

$$\mathrm{MAD}(z) = \mathrm{median}_{p\in G}\left|z_p - \mathrm{median}_{q\in G}(z_q)\right|,$$

$$\mathrm{RON}_{\mathrm{MAD}} = \frac{1.4826}{\sqrt{2}}\,\mathrm{MAD}(\Delta B).$$

Here q is just a second pixel index used to express the inner median. The factor 1.4826 puts MAD on the standard-deviation scale for a Gaussian distribution. The MAD estimate is reported separately; it does not replace ordinary variance in the conversion-gain calculation.

Conversion gain from flat and bias pairs

Form the raw flat-pair difference:

$$\Delta F_p = F_{1,p} - F_{2,p}.$$

The conversion gain is:

$$\boxed{g = \frac{\bar{F}_1 + \bar{F}_2 - \bar{B}_1 - \bar{B}_2}{v(\Delta F) - v(\Delta B)}} \quad [\text{electrons/ADU}].$$

The numerator is the sum of the two flat signals after removing the estimated bias levels. The denominator removes the readout-noise contribution from the flat-difference variance. The relation assumes photon shot noise, compatible flat illumination patterns and operation within the accepted linear signal range. Both numerator and denominator must be positive for a reported gain.

Use raw flat differences over the same accepted pixels, without scan correction, for this first photon-transfer estimator. Bias removal in the numerator uses the four image means. Changes in electronic structure between exposures remain an admission concern; this routine does not model or correct arbitrary drift.

Variance versus signal: which value represents the instrument?

There is no single instrument variance that applies at every signal level. Distinguish **readout variance at zero signal** from the **total variance at a specified signal**, and distinguish both from conversion gain.

For flat pair j, define the bias-subtracted average signal and half the difference variance:

$$\mu_j = \frac{\bar{F}_{j1} + \bar{F}_{j2} - \bar{B}_1 - \bar{B}_2}{2}, \qquad V_j = \frac{v(F_{j1} - F_{j2})}{2}.$$

Here j labels an integration-time pair, and subscripts $j1$ and $j2$ identify its two flat exposures. μ_j is in ADU and V_j is in squared ADU. The factor of two converts the difference variance to the variance of ONE exposure, assuming independent exposures with comparable noise statistics. For modest illumination mismatch it represents the average of their variances.

Define the independently measured raw-bias readout variance:

$$V_0 = \frac{v(\Delta B)}{2} = \mathrm{RON}^2_{\mathrm{ADU}}.$$

In the constant-gain, independent-pixel, photon-noise regime:

$$\boxed{V(\mu) = V_0 + \frac{\mu}{g_0}}.$$

The new symbol g_0 denotes a reference conversion gain in electrons/ADU, measured over an explicitly selected approximately linear signal interval. $V(\mu)$ denotes the single-exposure variance at bias-subtracted signal μ; V_0 is its zero-signal readout component.

If plotting the FULL difference variance instead, as in the owner's question:

$$v(\Delta F) = 2V_0 + \frac{2\mu}{g_0}.$$

Thus the slope of half-difference variance versus average signal is $1/g_0$; the slope of full-difference variance versus average signal is $2/g_0$. The ordinate is NOT the spatial variance of the average flat image.

Proposed reporting policy, for review:

- Use V_0 from the raw bias pair for readout variance. Zero signal is the reference; no illuminated flat point is selected to define readout noise.
- Estimate g_0 over a declared, verified approximately linear signal interval, above the region where read noise dominates and below significant curvature or saturation. Use a range of accepted pairs, not one arbitrarily chosen point. Record the interval, pair membership, fitting method and uncertainty; no numerical interval or tolerance is selected in this proposal.
- Keep the individual pair estimates versus signal:

$$g_{\mathrm{pair},j} = \frac{\mu_j}{V_j - V_0}.$$

In a curved relation this is a signal-dependent effective estimate. It need not equal a gain inferred from the local tangent. A local-slope diagnostic, if later implemented with a reviewed curve model, would be

$$g_{\mathrm{slope}}(\mu) = \left(\frac{dV}{d\mu}\right)^{-1}.$$

Both quantities agree with g_0 for a straight photon-transfer relation; they generally differ for a curved one. Neither alone proves an intrinsic electronics gain change.

- For the variance needed at a particular science signal, evaluate the validated variance-versus-signal model at that signal. Do not assign V_0 or the variance at a calibration flat level to every science pixel.
- If no sufficiently constrained approximately linear interval exists, report the reference gain as undetermined and retain the measured curve and pair estimates. Do not force a constant gain or extrapolate from saturation.

The intended first helper still returns each pair's measured quantities. A fitted reference gain/curve belongs to subsequent sequence-level analysis; this reporting proposal does not silently add a new curve fitter to that helper.

Curvature can reflect nonlinear response, changing pixel correlations/charge redistribution, illumination mismatch or other signal-dependent effects. Consequently, a straight-line fit across the whole curve cannot establish a universal gain, and a per-pixel variance alone does not describe covariance. The later application of a linearity correction must propagate the noise through that correction, as required by the controlling design.

As a methodological reference, ESO's VIMOS gain documentation describes paired estimates and photon-transfer fitting, and records a later restriction to low-signal flats. That instrument-specific threshold is not adopted for OmegaCAM.

Median centering and controlled lamps

Let $D_p = \Delta F_p$ and let m be its median over G:

$$m = \mathrm{median}_{p\in G}(D_p), \qquad D'_p = D_p - m.$$

Subtracting this one constant makes the median of the difference zero, but:

$$v(D') = v(D), \qquad \mathrm{MAD}(D') = \mathrm{MAD}(D).$$

Thus your proposed centering changes the displayed zero point, not the measured variance or MAD. No extra centering estimator or input mutation is needed. Retain the original difference median and the two

bias-subtracted flat levels as pair-consistency diagnostics. A multiplicative lamp change can leave a spatial response pattern after median centering; no automatic rescaling is added. The original uncentered levels remain available for the later linearity fit.

Proposed Eclipse interface

This is the C entry point for the measurements defined above:

```
int detector_gain_pair_checked(
    const image_t *flat_first, const image_t *flat_second,
    const image_t *bias_first, const image_t *bias_second,
    const pixelmap *selection, const int *zone,
    int with_mad, detector_gain_pair_result *out);
```

- The four image arguments supply the flat pair and the raw bias pair.
- `selection` is the optional common good-pixel map in Eclipse's existing convention: 1 means accepted, 0 means excluded.
- `zone` supplies the ROI boundaries, inclusive and counted from 1.
- `with_mad` requests the additional robust diagnostic.
- `out` receives the measurements, counts and availability/reason fields.

All four inputs have the same detector/readout, binning, dimensions and raw-to-active mapping. The recipe uses the existing crop operation to provide active-pixel images without scan subtraction. The selection is an explicit binary Eclipse view, not the eight-bit reason array.

The result retains the four image-level statistics, both difference statistics, the common valid/excluded counts, bias-subtracted pair signals, pair-mismatch diagnostics, conversion gain and raw-pair RON.

Errors, ownership and callers

Inputs/maps/ROI are borrowed and immutable; output is caller-owned and has no owned image allocations. Temporary images/maps use existing Eclipse allocation, cleanup and error conventions. Invalid dimensions/maps/ROI, invalid options, allocation failure or nonfinite arithmetic return an error without changing *out. Fewer than two common samples or nonpositive gain numerator/denominator produce an explicit unavailable measurement, never a nominal zero gain. RON can remain available when gain is unavailable. Reject aliasing/invalid output arguments as specified in the matching header documentation.

The numerical routine reports measurements, not scientific acceptance. The recipe applies explicit illumination-mismatch/range thresholds and records the decision. It calls the helper once per configured amplifier/ROI/pair and records detector identity, sample population, raw origin and estimator. Overlapping ROI estimates share pixels and are not independent observations.

No new global model fitter, default mismatch tolerance, defect criterion or uncertainty approximation is approved by this interface review.

A small synthetic example for checking the arithmetic

"Four-pixel fixture" meant **four artificial images, each containing only four pixels**. It did not mean four real OCAM exposures or a recommended measurement area. The four pixel positions are the same in every artificial image. All sixteen values below are invented for a simple software test.

Pixel position p	First flat $F_{1,p}$	Second flat $F_{2,p}$	First bias $B_{1,p}$	Second bias $B_{2,p}$
1	101	99	11	10
2	99	101	9	10
3	100	100	10	10
4	100	100	10	10

All entries are in ADU. All four positions are accepted, so $N = 4$. The four image means are:

$$\bar{F}_1 = \bar{F}_2 = 100 \text{ ADU}, \qquad \bar{B}_1 = \bar{B}_2 = 10 \text{ ADU}.$$

The pair differences, in pixel order, are:

$$\Delta F = (2, -2, 0, 0)\ \text{ADU}, \qquad \Delta B = (1, -1, 0, 0)\ \text{ADU}.$$

Both difference means are zero. With the sample-variance denominator $N - 1 = 3$:

$$v(\Delta F) = \frac{2^2 + (-2)^2 + 0^2 + 0^2}{3} = \frac{8}{3}\ \text{ADU}^2,$$

$$v(\Delta B) = \frac{1^2 + (-1)^2 + 0^2 + 0^2}{3} = \frac{2}{3}\ \text{ADU}^2.$$

The expected software-test results are therefore:

$$g = \frac{100 + 100 - 10 - 10}{8/3 - 2/3} = 90\ \text{electrons/ADU},$$

$$\text{RON}_{\text{ADU}} = \sqrt{\frac{2/3}{2}} = \sqrt{\frac{1}{3}}\ \text{ADU} \simeq 0.57735\ \text{ADU}.$$

These deliberately simple numbers make errors in subtraction, the $N - 1$ denominator or the factor of two easy to detect. **They are not expected OCAM gain/noise values and do not validate the physical method on real observations.**

Focused acceptance to prepare after interface review

Before execution, announce the inputs, functions, actions and pass criteria.

1. Recover the independently calculated results in the synthetic example above.
2. Verify that BPM exclusions and nonfinite samples select exactly the same population for all means and both variances. Check ROI coordinates and counts.
3. Swap the members of either pair: gain and RON must be unchanged, while the corresponding difference median changes sign. Verify separately that subtracting a constant leaves variance and MAD unchanged.
4. Check constant images, too few accepted pixels, nonpositive gain denominators and invalid maps. Report unavailable values or errors as appropriate; preserve caller output on error and never modify the inputs.
5. Exercise the new call through the existing operation template using synthetic pairs whose members are not consecutive. Check failure handling and cleanup.
6. Verify the factor-of-two convention with a synthetic constant-gain relation. Curved examples must retain signal-dependent pair results rather than being silently presented as one constant gain. Sequence-level fitting remains later.
7. Run focused Mac C17 checks first and validate the changed path on Chapman afterward. Do not repeat unchanged qualification matrices or alter production installations or original data.

Connection to the full linearity design

The earlier formula guide (PDF) and LaTeX source remain the controlling design. This increment implements only its paired gain/RON part.

The later shared shutter and linearity fit retains:

$$Q_a(s) = s + c_{2,a}s^2 + c_{3,a}s^3, \qquad Q_a(s_{rj}) = L_r(t_j + \delta).$$

Symbol	Meaning	Unit
a	Amplifier, or pixel in the later per-pixel mode	Identifier
r	Analysis region	Identifier
j	Exposure	Index

Symbol	Meaning	Unit
s	Bias-subtracted measured signal	ADU
s_{rj}	Signal for region r and exposure j, subject to the guide's pixel-model/aggregation rule	ADU
$Q_a(s)$	Linearized signal for model a	ADU
$c_{2,a}$	Quadratic coefficient of the inverse response	ADU^{-1}
$c_{3,a}$	Cubic coefficient of the inverse response	ADU^{-2}
L_r	Constant linearized illumination rate in region r	ADU/s
t_j	Exposure duration established from the declared input-time convention	s
δ	One signed duration offset shared by the mosaic; positive means longer exposure	s

The effective duration is $t_j+\delta > 0$. Keep the guide's monotonicity, validated signal bounds, rank, conditioning, covariance and aggregation requirements. No independent per-chip shutter offsets or unrestricted lamp-drift parameters are introduced. The exact joint numerical interface is a later review.

Selected Mac OmegaCAM inputs

Root: `/Users/fselman/Pictures/OCAM/bpm`. Owner-provided listing; files/pixels have NOT yet been independently inspected on the Mac. Ten BIAS/r_SDSS exposures precede twenty FLAT,DOME,DETGAIN/v_STRM exposures on 2017-03-13. Different bias/flat filters are recorded; detector/readout compatibility still needs header confirmation.

Bias file timestamps (`OMEGA.2017-03-13T<timestamp>.fits`): 09:50:05.624, 09:50:48.373, 09:51:31.403, 09:52:14.404, 09:52:57.394, 09:53:40.384, 09:54:23.384, 09:55:06.394, 09:55:49.384, 09:56:31.804.

Pair	Time (s)	First flat timestamp	Second flat timestamp	Chronological flat indices
P02	2	11:34:43.759	11:58:45.683	1,20
P70	70	11:35:30.570	11:56:50.383	2,19
P40	40	11:37:25.420	11:55:25.563	3,18
P05	5	11:38:50.390	11:54:35.393	4,17
P10	10	11:39:40.380	11:53:40.483	5,16
P30	30	11:40:35.390	11:52:25.393	6,15
P20	20	11:41:50.391	11:51:20.392	7,14
P50	50	11:42:55.391	11:49:45.392	8,13
P60	60	11:44:30.391	11:48:00.392	9,12
P07	7	11:46:16.391	11:47:08.392	10,11

This sequence has ten levels, not the design's representative 22-pair annual sequence. Do not reject it solely on that count or promise all products: admit each requested model using its actual rank, usable signal span, saturation and illumination evidence. Preserve header times; do not assume they are commanded times or already shutter-corrected.